LAB 8: SERVER CONFIGURATION – DHCP, DNS AND WEB SERVER IN CISCO PACKET TRACER

OBJECTIVES:

- To set up a server in Cisco Packet Tracer that provides DHCP, DNS, and HTTP services.
- To observe how client devices automatically receive IP configuration using DHCP.
- To understand how DNS converts domain names into IP addresses.
- To test web access from client PCs using a configured domain name.

THEORY:

In modern computer networks, servers play a major role in providing centralized services to clients. Services such as automatic IP assignment, name resolution, and web hosting are essential for smooth network communication. In this lab, a server is configured to deliver three important services — DHCP, DNS, and HTTP — within a small LAN environment using Cisco Packet Tracer.

1.Dynamic Host Configuration Protocol (DHCP):

DHCP is a protocol that automatically provides IP configuration details to network devices. These details include:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server Address

Without DHCP, administrators would need to manually configure each device, which increases errors and IP conflicts. When DHCP is enabled on a server in Packet Tracer, client PCs can automatically obtain network settings.

2.Domain Name System (DNS)

DNS is responsible for translating human-readable domain names (e.g., www.example.com) into machine-readable IP addresses. Without DNS, users would need to remember numerical IP addresses to access resources. In Packet Tracer, configuring DNS on a server allows client devices to resolve domain names to the correct IP address of the web server. This simulates real-world internet behavior, where

DNS servers act as the "phonebook" of the network.

3.Web Server (HTTP Service) :

A web server hosts websites and responds to client requests using the Hypertext Transfer Protocol (HTTP). When a client enters a domain name in its browser, the DNS resolves it to the server's IP, and the HTTP service delivers the requested web page. In Packet Tracer, enabling the HTTP service on the server allows PCs to access and display web content, verifying that DHCP and DNS are functioning correctly.

4.Integration of Services:

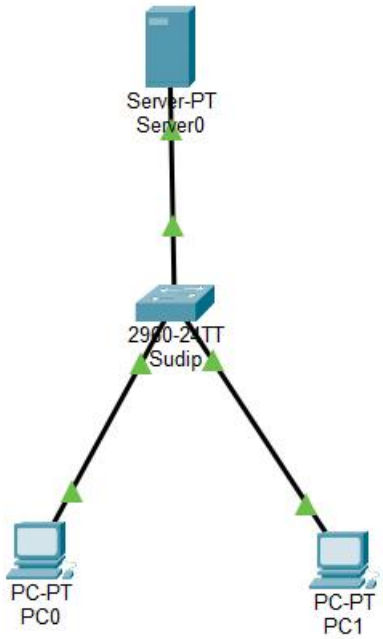
The three services operate together as follows:

- DHCP assigns IP configuration to PCs.
- DNS resolves the domain name to the server's IP.
- HTTP service delivers the webpage.

This integration simulates how real-world enterprise networks function.

NETWORK TOPOLOGY

This network topology illustrates a basic star configuration where all devices connect through a central switch. The server provides DHCP, DNS, and Web services, while the two PCs act as clients to test connectivity and service functionality. This setup demonstrates how centralized resources can efficiently manage IP addressing, domain resolution, and web hosting in a small network.



CONFIGURATION:

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PCs,

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Note: The PCs IP address are dynamically generated via DHCP server. The above mentioned is obtained from DHCP server (no manual registering).

SERVER CONFIGURATION
Step 1 : Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:
   - IP Address: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
   - DNS Server: 192.168.1.2

DHCP Configuration :
Step 2 : Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP: ON

Step 3 : Create DHCP Pool

Field	Value
Pool Name	LAB_Pool
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add
Verification :
On each PC:

- Go to Desktop → IP Configurationzax
- Select DHCP

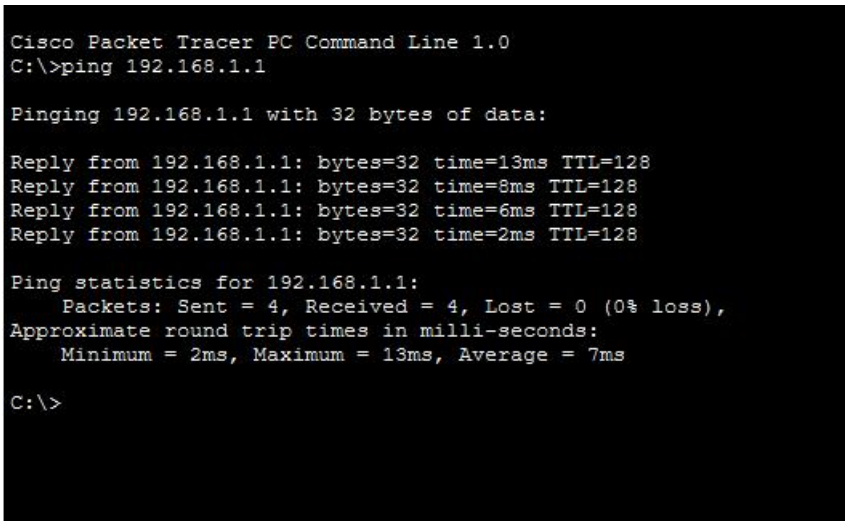


Fig: Ping Test from PC1 TO PC0

DNS Server Configuration :
Step 4 : Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5 : Add DNS Records

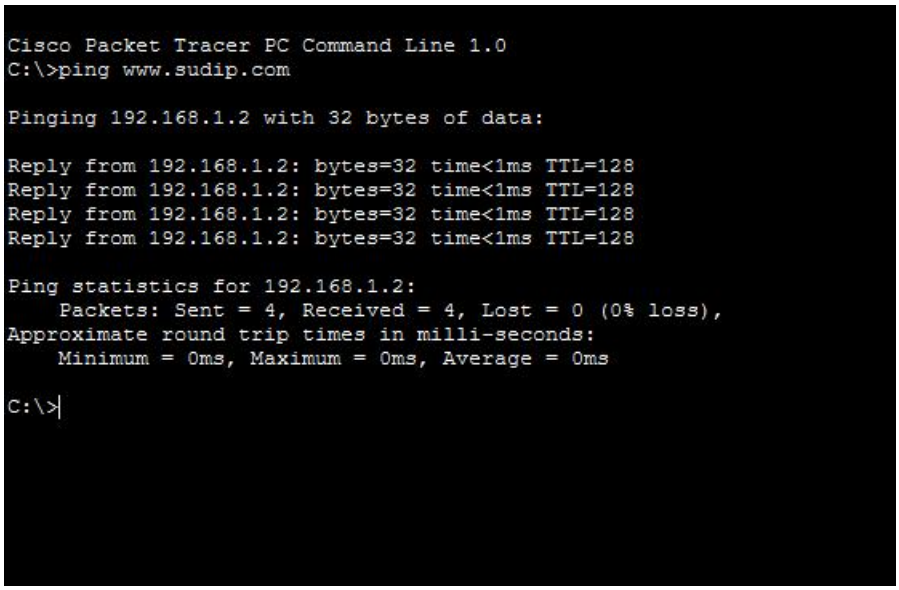
- Name: www.sudip.com
- Type: A record
- Address: 192.168.1.2

Click Add

Verification :

On each PC:

- Open Command Prompt
- Ping: www.sudip.com



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Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.sudip.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
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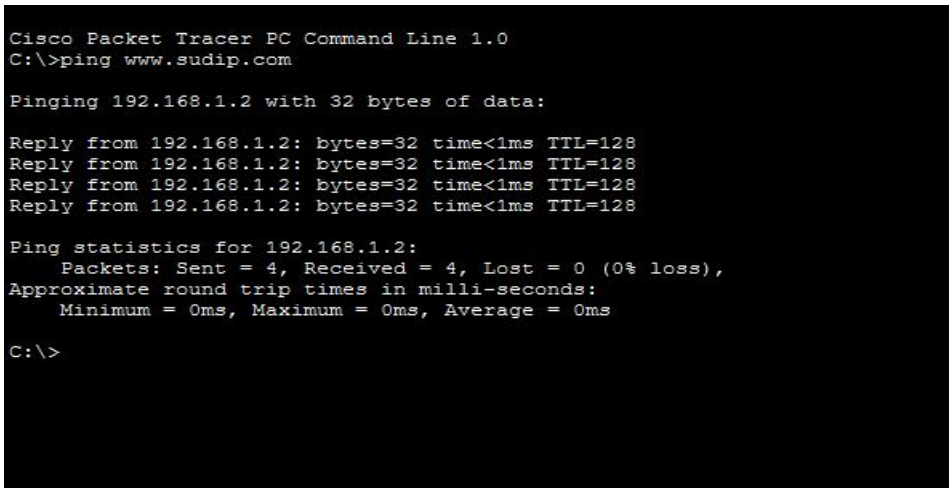
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig: Ping Test from PC0



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C:\>
```

Fig: Ping Test from PC1

Web Server Configuration :

Step 6 : Enable HTTP Service:

1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)

Verification :

On PC:

- Go to Desktop → Browser
- Type: http://www.sudip.com

DISCUSSION:

In this lab, a server was configured to provide DHCP, DNS, and Web services within a star topology network using Cisco Packet Tracer. The DHCP service automatically assigned IP addresses to client computers, reducing the need for manual configuration and minimizing errors. The DNS service translated domain names into the corresponding IP address of the server, allowing users to access the website easily through a web browser. The web server successfully hosted and delivered web content to connected clients. Overall, the lab demonstrated how integrating multiple services on a single server improves network efficiency, simplifies management, and ensures smooth communication among devices.

CONCLUSION:

The lab demonstrated that DHCP, DNS, and Web services can be properly configured and verified in Cisco Packet Tracer. DHCP automated the IP address allocation process, DNS enabled proper name resolution, and the web server ensured successful content delivery to users. This practical implementation highlighted the importance of centralized server configuration in creating a reliable, efficient, and user-friendly network environment.